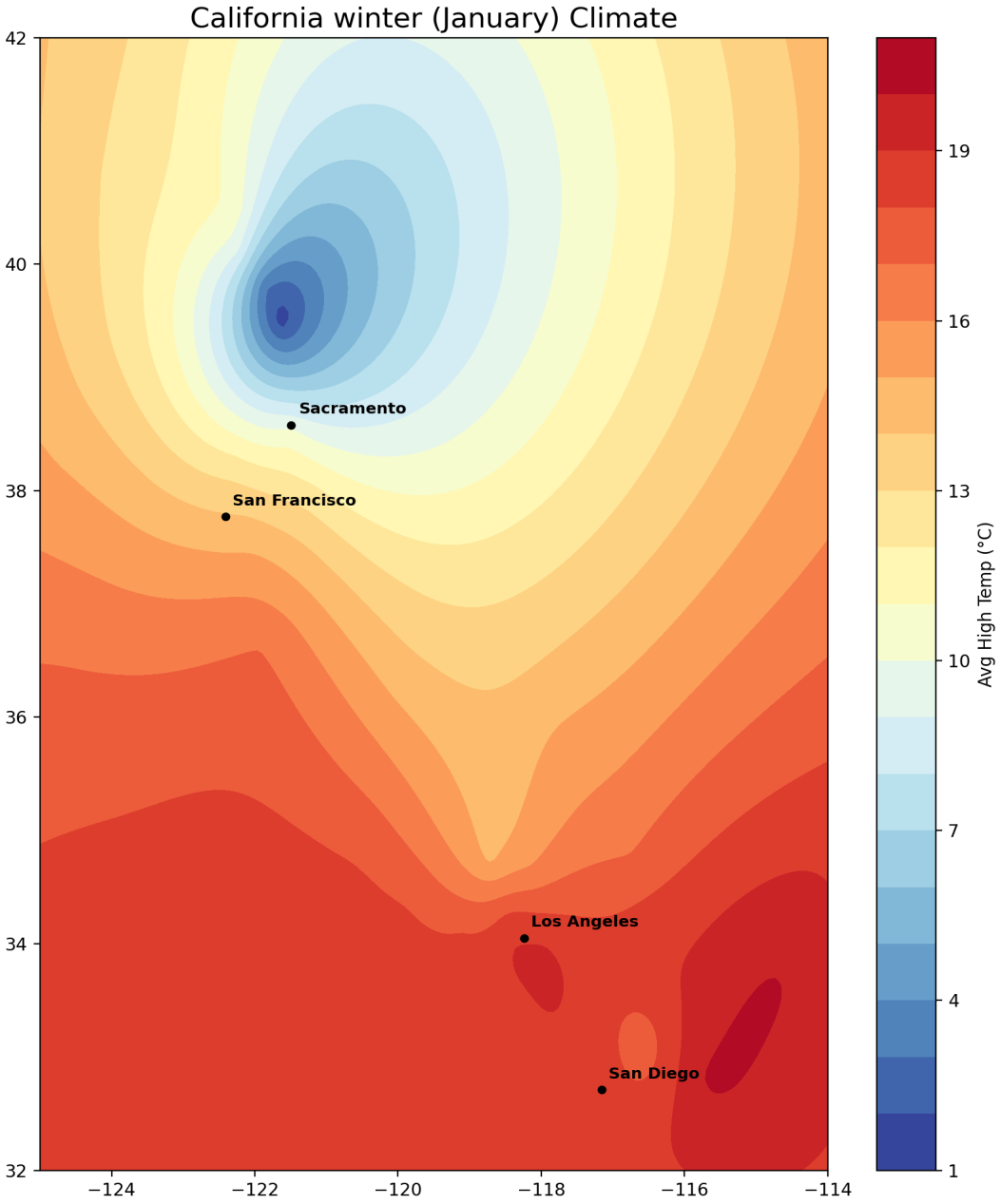
California Residential Climate Analysis

Prepared for Real Estate Investors and Homebuyers

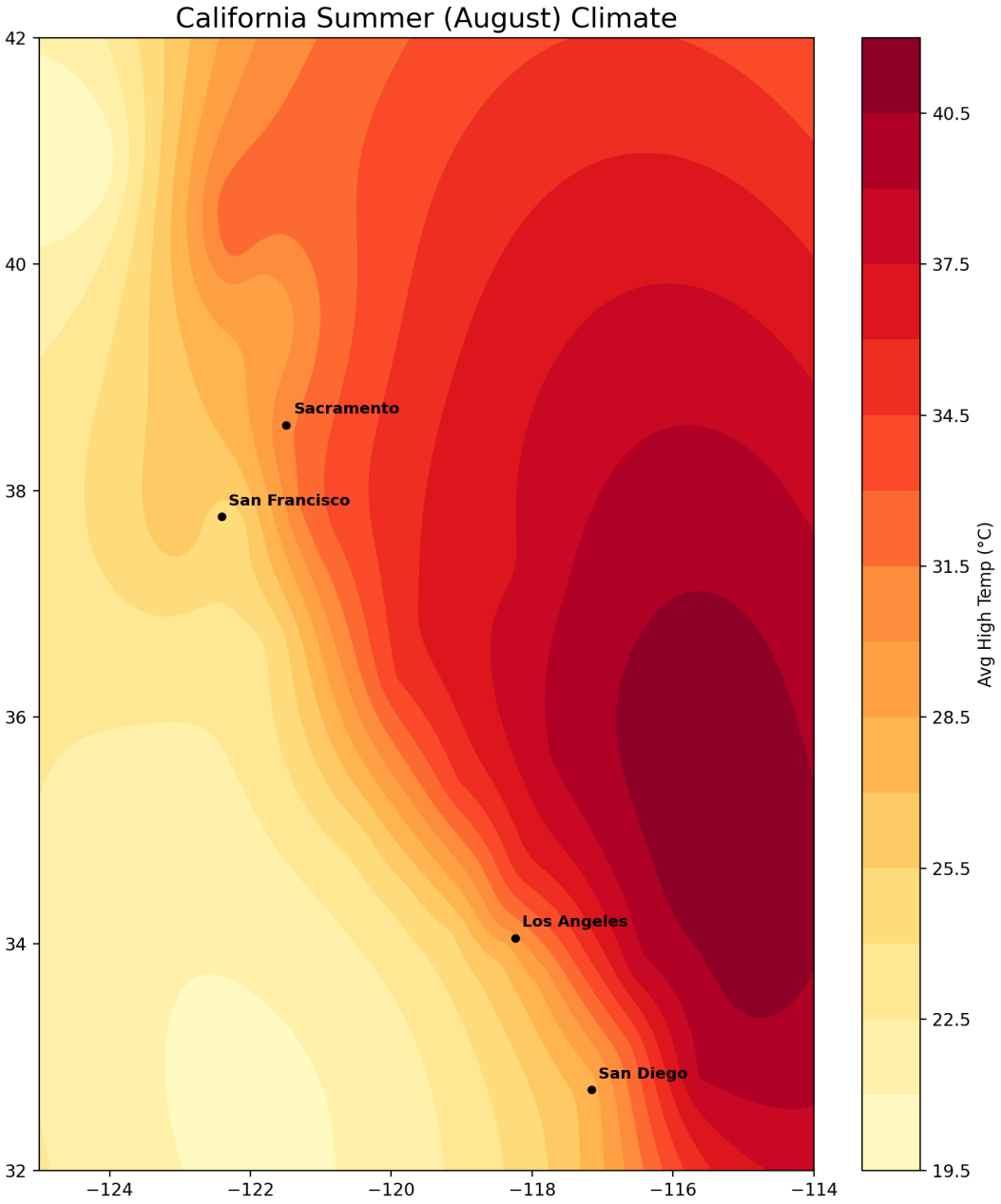
Executive Summary

This report provides an in-depth look at seasonal temperature variations across California using station normals, Kriging interpolation, and city-level sampling. The goal is to translate climate surfaces into practical residential location guidance for investors and homebuyers.

1. Winter Climate (January)



2. Summer Climate (August)



3. Warmest and Coolest Cities

January Top 10 Warmest Cities

City Name	Temp (°C)
Imperial	20.1
El Centro	20.1
Holtville	20.1
Heber	20.1
Mesa Verde	20.1
Brawley	20.0
Palo Verde	20.0
El Centro Naval Air Facility	20.0
Seeley	20.0
Ripley	20.0

January Top 10 Coolest Cities

City Name	Temp (°C)
Thermalito	1.6
Oroville	2.0
Butte Valley	2.0
Cherokee	2.2
Palermo	2.2
South Oroville	2.2
Kelly Ridge	2.3
Oroville East	2.3
Richvale	2.3
Butte Creek Canyon	2.4

August Top 10 Warmest and Coolest Cities

August Top 10 Warmest Cities

City Name	Temp (°C)
Charleston View	41.2
Baker	41.1
Needles	41.1
Tecopa	41.0
Shoshone	41.0
Big River	40.9
Bluewater	40.9
Mesa Verde	40.9
Blythe	40.8
Ripley	40.8

August Top 10 Coolest Cities

City Name	Temp (°C)
Manila	19.8
Fairhaven	19.9
Samoa	19.9
Bayview	19.9
Eureka	19.9
McKinleyville	20.0
Trinidad	20.2
Westhaven-Moonstone	20.2
Fields Landing	20.2
Humboldt Hill	20.2

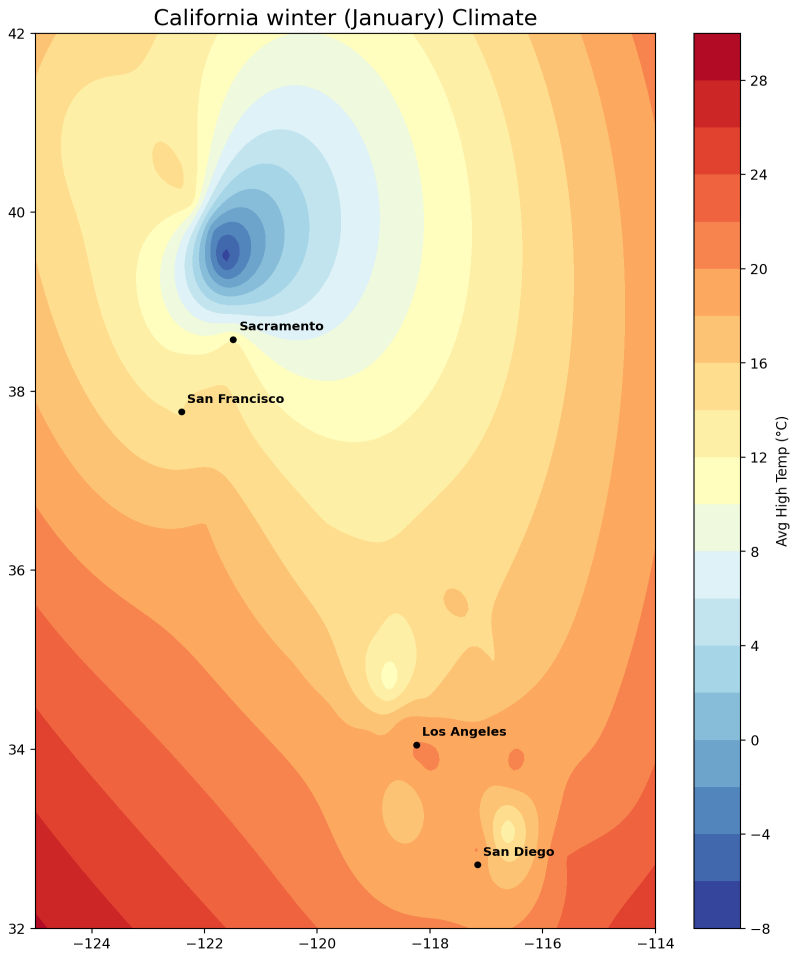
4. Major City Decision Summary

The table below samples the same interpolated climate surfaces at major California city locations. This keeps the map reading connected to familiar places instead of leaving the result as a color surface only.

City	January High (°C)	August High (°C)	Residential Climate Note
Los Angeles	18.8	30.2	Warm summer market with mild winter conditions.
San Francisco	14.1	25.0	Cooler coastal market with muted seasonal extremes.
Sacramento	9.7	31.2	Hotter inland summer profile with cooler winter mornings.
San Diego	18.8	27.8	Consistently mild coastal climate profile.

5. Technical Addendum: Interpolation Methodology

The primary analysis used **Ordinary Kriging**, a geostatistical method that accounts for both distance and the spatial autocorrelation of data points. For comparison, **IDW (Inverse Distance Weighting)** was also performed.



Results indicate that Kriging provides a smoother, more realistic transition of temperatures in mountainous regions compared to IDW, which can create 'bulls-eye' artifacts around isolated stations.